

① $n = 6$.

$$P(x > 3) = \frac{3}{6} = \frac{1}{2}$$

② $n = 52$

$P(\text{face card}) = ?$ no. of face cards = 12

$$P(FC) = \frac{12}{52} = \frac{3}{13}$$

③ $n(R) = 7$.

$n(B) = 5$.

$n(G) = 8$.

$n(T) = 20$.

$$P(R \cap B) = P(R) = \frac{7}{20}$$

④

$n(T) = \{4, 8, 12, 16, 20\} \rightarrow 5$

$$P(\text{divisible by 4}) = \frac{5}{20} = \frac{1}{4}$$

⑤

$$n = 50.$$

$$n(M) = 28$$

$$n(S) = 22$$

$$\cancel{n(M)} = n(M \cap S) = 10$$

$$n(M \cup S) = ?$$

$$n(M \cup S) = 28 + 22 - 10 = 40.$$

$$P(M \cup S) = \frac{40}{50} = \frac{4}{5}.$$

⑥

$$n = 52.$$

$$n(H) = 13.$$

$$n(K) = 4.$$

$$P = \cancel{52} P(H) + P(K) - 1$$

$$= \frac{13 + 4 - 1}{52} = \frac{16}{52} = \frac{4}{13}.$$

⑦

$$n = 6.$$

$$P(2 \cup 5) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6}.$$

⑧

$$n = 52$$

$$n(Q) = 4$$

$$n(J) = 4$$

$$P(Q \cup J) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}.$$

9

$$n(n) = 36$$

$$n(T) = 4$$

$$n(2H) = \frac{1}{4}$$

10

$$P(4H) = \frac{1}{12}$$

	1	2	3	4	5	6
T						
H				4H		

$$P(4 \text{ on die}) = \frac{1}{6} \quad P(4 \text{ a head}) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$

$$P(4H) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$$

11

$$P(m) = 0.9$$

$$P(am) = 0.6$$

$$P(m \cap am) = 0.9 \times 0.6 = 0.72$$

12

$$P(p) = 0.7$$

$$P(c) = 0.6$$

$$\overline{P(p)} = 1 - 0.7 = 0.3$$

$$\overline{P(c)} = 1 - 0.6 = 0.4$$

$$P(\text{at least 1}) = 1 - 0.12 = 0.88$$

13.

$$n(W) = 5.$$

$$n(B) = 3.$$

$$P(\text{both are white}) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56}$$

14

$$n = 52.$$

aces.

without replacement.

$$P = \frac{4}{52} \times \frac{3}{51} = \frac{12}{2652} = \frac{1}{221}$$

15.

$$n(R) = 4 \quad n(B) = 6$$

$$P = \frac{4}{10} \times \frac{6}{9} = \frac{24}{90} = \frac{4}{15}$$

16

$$n(B) = 5.$$

$$n(G) = 4.$$

$$P(\text{all girls}) = \frac{4}{9} \times \frac{3}{4} = \frac{12}{36} = \frac{1}{3}$$

17. $n(7) = 52$

$n(R) = 26$

$n(K) = 2$

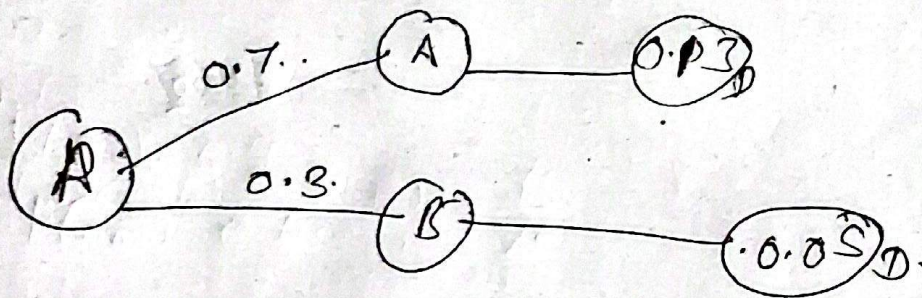
$$P(K) = \frac{n(K)}{n(R)} = \frac{2}{26} = \frac{1}{13}$$

18. $n = 6$

$\{0, 2, 4, 6\}$

$P(>4) = \frac{2}{6} = \frac{1}{3}$

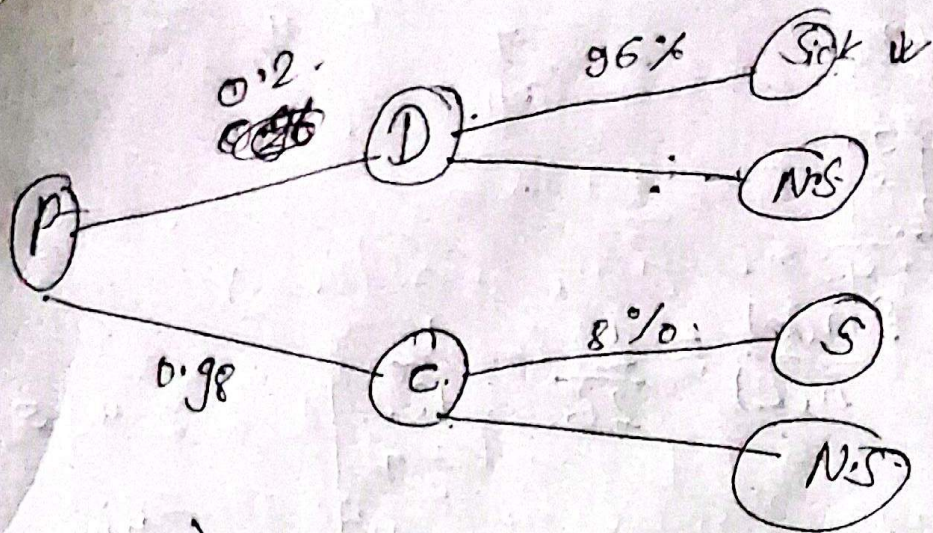
19.



$$P\left(\frac{B}{D}\right) = \frac{P\left(\frac{D}{B}\right) \times P(B)}{P(D)} = \frac{0.05 \times 0.3}{0.7 \times 0.3 + 0.3 \times 0.05}$$

$$= \frac{0.015}{0.036} = \frac{5}{12} = 0.4167$$

20



$$P(D/C) = P(C/D) \times$$

$$P(D) = \frac{P(C/D) \times P(D)}{P(C/D) \times P(D) + P(D/C) \times P(C)} = \frac{0.96 \times 0.02}{0.96 \times 0.02 + 0.98 \times 0.98} = 0.1967$$